

Vinicius Santana

Software Engineer | Backend & Full Stack | Python, Java, TypeScript, AWS

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TECHNICAL SKILLS

Languages: Python, Java, TypeScript, Go

Backend: FastAPI, Spring Boot, OpenAPI, SQLAlchemy, Pydantic

Frontend: React, TanStack Query

Databases: PostgreSQL, Redis, DynamoDB

Cloud & DevOps: AWS, Docker, Kubernetes (k8s), GitHub Actions, Jenkins, Ansible, Grafana, Prometheus

Testing: Pytest, Testcontainers, Hypothesis, Vitest, Playwright, CI/CD

Architecture: Hexagonal Architecture, Event-Driven Design, Multi-Tenancy

Spoken Languages: Portuguese (Native), English (C1, CEFR)

PROFESSIONAL EXPERIENCE

Software Engineer – Health Data Systems

Nov 2023 – Present

Municipality of Presidente Epitácio, Brazil | Official title: Administrative Clerk

- Reduced healthcare claim rejection rate from 12%+ to under 1% for 8 consecutive billing cycles by implementing 11 auditable Python/SQL validation rules that reconcile federal (CNES, SIA/SIH) and local records.
- Cut monthly manual review by 95% (21,000 to ~1,000 records across 17 facilities) by building a validation and reconciliation engine with FastAPI, Spring Boot and PostgreSQL.
- Reduced workforce reconciliation from 240+ to ~4 person-hours per cycle (-98%) by replacing spreadsheet comparisons with an automated ELT pipeline with data-provenance controls.
- Cleared a backlog of 10,000+ billing records (100% of pending submissions) in 3 months by standardising validation, correction and resubmission workflows across 4 legacy systems and 3 partner hospitals.
- Resolved performance degradation in a production Spring Boot/PostgreSQL service, confirmed against instrumented latency baselines, by fixing identifier lifecycle handling and eliminating N+1 queries.
- Cut mandatory dengue reporting from 7 manual steps to 1 checkbox by shipping a Java/Spring clinical-workflow feature.
- Restored a 9 GB PostgreSQL-backed application in under 1 hour after a failed update, and automated provisioning of 2 production replicas serving 348 professionals, using Ansible, Jenkins, Docker and WAL-G backups.

IT Infrastructure & Systems Administration Intern

Nov 2021 – Oct 2023

Municipality of Presidente Epitácio, Brazil

- Maintained 99.9% availability for services used by 43,000+ residents and eliminated 100% of virtualisation licensing costs by migrating workloads to a high-availability Proxmox/Ceph cluster on Linux.
- Rerouted traffic from unhealthy nodes within 1–3 min by introducing Nginx/HAProxy health-check failover and Prometheus monitoring.

PROJECTS

CnesData – Multi-Tenant Registry Reconciliation Platform

2024 – Present

React, TypeScript, FastAPI, Go, PostgreSQL, AWS | cnesdata.com.br

- Onboarding 3 municipal customers via supervised rollouts by building a multi-tenant React/TypeScript dashboard and FastAPI services with PostgreSQL row-level security.
- Enabled municipalities to share data without exposing internal databases by designing Go edge agents and a hybrid local/cloud architecture (S3/Parquet, Athena, DynamoDB).
- Made ingestion retry-safe under distributed load by implementing a lease-based async job queue with idempotent upserts, exponential backoff, circuit breakers and a dead-letter queue.
- Blocked regressions before merge by enforcing CI gates for unit, integration, property-based and E2E tests (Pytest, Testcontainers, Vitest, Playwright).

LimnoPulse – Event-Driven Monitoring & Alerting Platform

2026 – Present

Python, FastAPI, Go, DynamoDB, Redis, AWS SQS/SES | limnopulse.com

- Delivered real-time alerts for 8 live sensors by building tenant-scoped FastAPI APIs with role-based authorisation and Go notification workers (email/Telegram) on SQS and SES.
- Guaranteed consistent incident state under retries and duplicate delivery (covered by concurrency tests) by implementing optimistic locking, idempotent writes and a durable outbox.

EDUCATION

BSc in Computer Science – Universidade Paulista (UNIP), Brazil | GPA 8.85/10

2021 – 2026

Technical Diploma in Systems Development – IFSP, Brazil | GPA 8.87/10

2018 – 2021